



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

SEPTEMBER 2003

YEARLY EXAMINATION

YEAR 11

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes.
- Working time 2 hours.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for messy or badly arranged work.
- Start each **NEW** section in a separate answer booklet.

Total Marks - 120 Marks

- Attempt Questions 1 - 4
- All questions are NOT of equal value.

Examiner: *E. Choy*

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Attempt Questions 1 – 4
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Answer each question in a SEPARATE writing booklet.

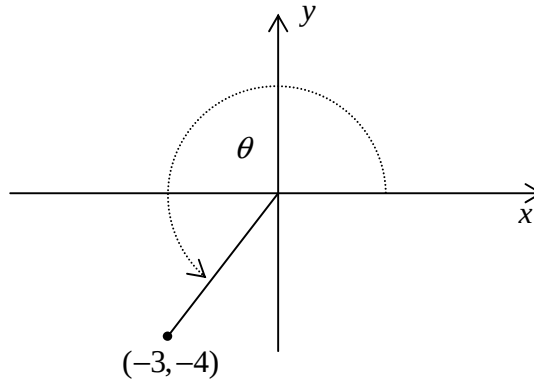
Question 1 (35 marks) Use a SEPARATE writing booklet		Marks
(a)	Find the value of x given that $x^3 = 3^6$.	2
(b)	Solve $(x-2)(x+1) \geq 0$.	2
(c)	Sketch $y = x+1 $, showing the x and y intercepts.	2
(d)	If $\sec \theta = 2$, find the possible values of $\tan \theta$.	2
(e)	Find, in EXACT general form, the equation of the line that passes through the point $(-1, 3)$ and has an angle of inclination, to the positive direction of the x axis, of 150° .	3
(f)	If $f(x) = \frac{x}{1-3x}$, find and simplify $f\left(\frac{1}{x}\right)$.	2
(g)	Solve the inequation $ x+1 < 2$	2
(h)	If $y = 8\sqrt{x}$, find $\frac{dy}{dx}$.	2
(i)	Complete the square to find the minimum value of the quadratic function $y = x^2 + 4x - 6$.	2

Question 1 continued

Marks

- (j) Write down the value of $\sin \theta$ in the diagram below.

2



- (k) Solve $2 \sin 2\theta = -1$ where $0 \leq \theta \leq 2\pi$.

3

- (l) If $(x-1)$ is a factor of $p(x) = x^3 + ax^2 - 2x - 4$, find the value(s) of a .

2

- (m) Simplify $\frac{\log_2 32}{\log_2 16}$

3

- (n) Find A and B if $x^3 - 27 = (x-3)(x^2 + Ax + B)$.

3

- (o) If $f(x) = x^2 - 4$ and $g(x) = x - 2$, find in simplest form $f(g(x))$.

3

Question 2 (32 marks) Use a SEPARATE writing booklet

Marks

- (a) If $p = 1 + \sqrt{2}$ and $q = 1 - \sqrt{2}$ find
- (i) $p - q$ 2
- (ii) pq 2
- (b) Differentiate the following with respect to x :
- (i) $y = 4x^5 + 2x^2 - 1$ 2
- (ii) $y = \frac{7}{x}$ 2
- (iii) $y = (4x^2 - 3x)^{12}$ 3
- (c) The limiting sum of the geometric series $a + \frac{a}{2} + \frac{a}{4} + \dots$ is 6. 3
Find the value of a .
- (d) Consider the function $f(x) = x^2 + 3x$.

Show that $f(x+h) - f(x) = 2xh + h^2 + 3h$ 2
- (e) (i) Show that there are 21 terms in the arithmetic series 2
 $-2 + 1 + 4 + \dots + 58$.
- (ii) Hence, or otherwise, find the sum of the 21 terms. 1
- (f) Find an equation in terms of x and y that is independent of θ 2
$$\left. \begin{array}{l} x = 2 \cos \theta \\ y = \sin \theta \end{array} \right\}$$
- (g) (i) Show that $\frac{1 - \cos \theta}{\sin \theta} = \tan \frac{\theta}{2}$ 3
- (ii) Hence find the exact value of $\tan 15^\circ$. 2
- (h) A and B are points $(-5, 1)$ and $(2, 2)$ respectively. Find the 3
coordinates of the point which divides A and B **externally** in
the ratio 3 : 2.
- (i) If $a + b = 1$, show that $(a^2 - b^2)^2 + ab = a^3 + b^3$ 3

Question 3 (20 marks) Use a SEPARATE writing booklet			Marks
(a)	(i)	Express $\sqrt{3} \sin x - \cos x$ in the form $R \sin(x - \alpha)$, where $R > 0$.	2
	(ii)	Hence, solve the equation $\sqrt{3} \sin x - \cos x = 1$, for $0 \leq x \leq 2\pi$.	2
(b)		Find the general solutions of $\sin x + \cos 2x = 1$.	2
(c)	(i)	Show that the equation of the normal to the curve $x^2 = 4y$ at the point $(2p, p^2)$ is $x + py = 2p + p^3$.	2
	(ii)	If the normal passes through the point $(-2, 5)$ find the values of p .	2
(d)		When the polynomial $P(x)$ is divided by $A(x) = (2x+1)(x-3)$, it gives a quotient $Q(x)$ and a remainder $R(x)$. Write the general form of $R(x)$. Justify your answer.	1
(e)	(i)	Show that the point $P(2, 7)$ lies on the line $2x - y + 3 = 0$	1
	(ii)	Hence find the distance between the parallel lines $2x - y + 3 = 0$ and $2x - y - 11 = 0$	2
(f)		$A(-2, -5)$ and $B(1, 4)$ are 2 points. Find the acute angle θ between the line joining A and B and the line $x + 2y + 1 = 0$, giving the answer correct to the nearest minute.	2

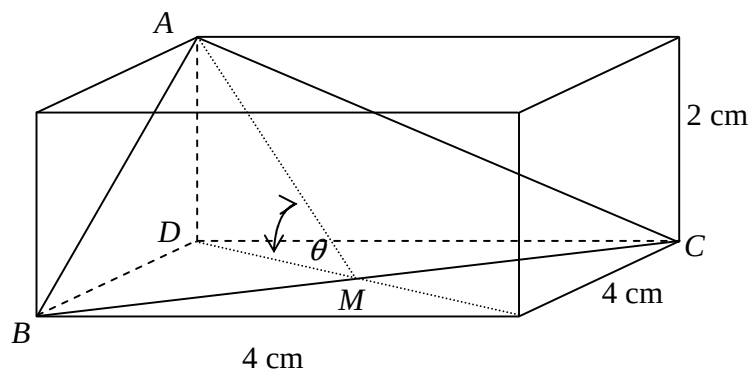
Question 3 is continued on the next page

Question 3 continued

Marks

- (g) The prism in the diagram below has a square base of side 4 cm and its height is 2 cm. ABC is a diagonal plane of the prism. Let θ be the acute angle between the diagonal plane and the base of the prism.

- (i) Show that $MD = 2\sqrt{2}$ cm. 2
- (ii) Hence find θ , correct to the nearest minute. 2



NOT TO SCALE

Question 4 (33 marks) Use a SEPARATE writing booklet

Marks

- (a) The quartic equation $x^4 - 4x^3 + 2x^2 - 3x + 2 = 0$ has roots α, β, γ and δ . Find the value of:

(i) $\alpha + \beta + \gamma + \delta$ 1

(ii) $\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta$ 1

(iii) $\alpha\beta\gamma\delta$ 1

(iv) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}$ 2

- (b) Given that $x = 1$ is a double root of the equation

$$6x^4 - 7x^3 + cx^2 + 13x - 4 = 0$$

(i) Show that $c = -8$ 2

(ii) Hence find the other roots. 2

- (c) If $\log_5 8 = a$, prove that $\log_{10} 2 = \frac{a}{a+3}$ 3

- (d) (i) By expanding $\cos(2A + A)$, show that 3

$$\cos 3A = 4\cos^3 A - 3\cos A$$

(ii) Hence show that if $2\cos A = x + \frac{1}{x}$, then $2\cos 3A = x^3 + \frac{1}{x^3}$ 2

Question 4 is continued on the next page.

Question 4 continued

Marks

- | | | | |
|-----|--|--|---|
| (e) | (i) | Show that the equation of the tangent at the point $P(2ap, ap^2)$ to $x^2 = 4ay$ is given by $y = px - ap^2$. | 2 |
| | (ii) | Write down the equation of the tangent at the point $Q(2aq, aq^2)$. | 1 |
| | (iii) | Find the coordinates of M the midpoint of chord PQ . | 1 |
| | (iv) | The tangents at P and Q meet at T . Find the coordinates of T . | 2 |
| | (v) | Show that TM is parallel to the axis of the parabola. | 2 |
| | (vi) | K is the midpoint of TM . Find the locus of K . | |
| (f) | <p>Three tangents to the parabola $x^2 = 4ay$ form a triangle PQR and the lines QR, RP and PQ make acute angles $\alpha_1, \alpha_2, \alpha_3$ respectively with the tangent at the vertex.</p> <p>If d_1, d_2 and d_3 are the respective distances of the focus from these tangents and if r_1, r_2 and r_3 are the respective distances of the focus from the vertices P, Q and R of ΔPQR, show that:</p> | | |
| | (i) | $d_1 \cos \alpha_1 = d_2 \cos \alpha_2 = d_3 \cos \alpha_3 = a$ | 2 |
| | (ii) | $d_1 r_1 = d_2 r_2 = d_3 r_3$ | 2 |
| | (iii) | $r_1 r_2 r_3 = \frac{d_1^2 d_2^2 d_3^2}{a^3}$ | 2 |

THIS IS THE END OF THE PAPER



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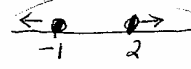
Mathematics Extension

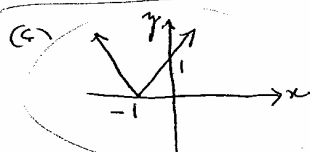
Sample Solutions

QUESTION 1.

(a) $x^3 = 3^6$
 $x = (3^6)^{\frac{1}{3}}$

$= 3^2$
 $x = 9$ ✓✓

(b) 
 $x < -1, x > 2$ ✓✓



(d) $\sec \theta = 2$

now $\sec \theta = 1 + \tan \theta$

$4 = 1 + \tan \theta$

$\tan \theta = 3$

$\tan \theta = \pm \sqrt{3}$ ✓✓

(e) $m = \tan 150^\circ$
 $= -\frac{1}{\sqrt{3}}$

$\therefore y - 3 = -\frac{1}{\sqrt{3}}(x + 1)$

$\sqrt{3}y - 3\sqrt{3} = -x - 1$

$x + \sqrt{3}y + 1 - 3\sqrt{3} = 0$ ✓✓

(f) $f(x) = \frac{x}{1-3x}$

$f\left(\frac{1}{2}\right) = \frac{\frac{1}{2}}{1-\frac{3}{2}}$

$= \frac{1}{x-3}$ ✓✓

(g) $|x+1| < 2$

$-2 < x+1 < 2$

$-3 < x < 1$ ✓✓

(h) $y = 8\sqrt{x}$
 $= 8x^{\frac{1}{2}}$

$\frac{dy}{dx} = 4x^{-\frac{1}{2}}$
 $= \frac{4}{\sqrt{x}}$ ✓✓

(i) $y = x^2 + 4x - 6$

$= x^2 + 4x + 4 - 6 - 4$

$= (x+2)^2 - 10$

$\therefore \text{MIN value is } -10$ ✓✓

(j) $\left(\frac{-4}{5}\right)$ ✓✓

$\left(\tan^{-1} \frac{3}{5}\right)$

(k) $2 \sin 2\theta = -1$ $0 \leq \theta \leq 2\pi$

$\sin 2\theta = -\frac{1}{2}$ $\therefore 0 \leq 2\theta \leq 4\pi$

$2\theta = \frac{7\pi}{6}, \frac{11\pi}{6}, \frac{19\pi}{6}, \frac{23\pi}{6}$

$\therefore \theta = \frac{7\pi}{12}, \frac{11\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12}$ ✓✓

(l) $P(x) = 0$

$\therefore 1 + a - 2 - 4 = 0$

$a - 5 = 0$

$a = 5$ ✓✓

(m) $\frac{\log_2 32}{\log_2 16} = \frac{\log_2 2^5}{\log_2 2^4} = \frac{5 \log_2 2}{4 \log_2 2} = \frac{5}{4}$ ✓✓

(n) $x^3 - 27 = (x-3)(x^2+3x+9)$

$\therefore x^2+3x+9 \equiv x^2+Ax+B$

$A=3$
 $B=9$ ✓✓

(o) $\log(x) = (x-2)^2 - 4 = x^2 - 4x$

Q2. a) $p_1 q = 2\sqrt{2}$ while ii) $p_1 q = -1$

b) i) $\frac{dy}{dx} = 20x^4 + 4x$ ii) $-\frac{1}{x^2}$ iii) $12(8x-3)(4x^2-3x)^4$

c) $b = \frac{a}{1-\frac{1}{2}}$
 $a = 3.$

d) $f(x+h) - f(x) = (x+h)^2 + 3(x+h) - x^2 - 3x$
 $= 2xh + h^2 + 3h$

e) i) $S_8 = -2 + (n-1) \times 3$ so $n = 21$

ii) $S_{21} = \frac{21}{2}(-2 + 58) = \frac{21}{2} \times 56 = 588$

f) $x = 2\cos\theta$ so $x^2 = 4\cos^2\theta$ and $x^2 = 4\cos^2\theta$
 $y = \sin\theta$ $y^2 = \sin^2\theta$ $4y^2 = 4\sin^2\theta$
 $4y^2 + x^2 = 4\cos^2\theta + 4\sin^2\theta$
 $x^2 + 4y^2 = 4$

g) i) $\frac{1-\cos\theta}{\sin\theta} = \tan\frac{\theta}{2} \Rightarrow \frac{1-\frac{1-t^2}{1+t^2}}{\frac{2t}{1+t^2}}$ where $t = \tan\frac{\theta}{2}$
 $= \frac{2t^2}{2t} = t = \tan\frac{\theta}{2}$

ii) $\tan 15^\circ = \frac{1-\cos 30^\circ}{\sin 30^\circ} = \frac{1-\frac{\sqrt{3}}{2}}{\frac{1}{2}} = 2-\sqrt{3}$

h) take $3x-2 \left(\frac{3x^2-2x-5}{1}, \frac{3x^2-2x+1}{1} \right) = (16, 4)$

i) $(a^2-b^2)^2 + ab$ a^3+b^3
 $(a+b)(a-b)^2 + ab$ while $=(a+b)(a^2-ab+b^2)$
 $(a-b)^2 + ab$ $= a^2-ab+b^2$
 a^2-ab+b^2 $= (a^2-b^2)^2 + ab = a^3+b^3$

Question 3

(a) (i) $\sqrt{3} \sin x - \cos x \equiv R \sin(x - \alpha)$
 $R \sin(x - \alpha) = R(\sin x \cos \alpha - \cos x \sin \alpha)$
 $= R \sin x \cos \alpha - R \cos x \sin \alpha$
 $= (R \cos \alpha) \sin x - (R \sin \alpha) \cos x$
 So $R \cos \alpha = \sqrt{3}$ —(1)
 $R \sin \alpha = 1$ —(2)

$$(2) \div (1) \Rightarrow \frac{\sin \alpha}{\cos \alpha} = \tan \alpha = \frac{1}{\sqrt{3}} \Rightarrow \alpha = 30^\circ \text{ or } \frac{\pi}{6}$$

$$(2)^2 + (1)^2 \Rightarrow R^2 \sin^2 \alpha + R^2 \cos^2 \alpha = 4$$

$$\therefore R^2 (\sin^2 \alpha + \cos^2 \alpha) = 4 \Rightarrow R^2 = 4$$

$$\therefore R = 2$$

(ii) $\sqrt{3} \sin x - \cos x = 1 \Rightarrow 2 \sin \left(x - \frac{\pi}{6} \right) = 1$

$$\sin \left(x - \frac{\pi}{6} \right) = \frac{1}{2}$$

$$0 \leq x \leq 2\pi \Rightarrow 0 - \frac{\pi}{6} \leq x - \frac{\pi}{6} \leq 2\pi - \frac{\pi}{6}$$

$$-\frac{\pi}{6} \leq x - \frac{\pi}{6} \leq \frac{11\pi}{6}$$

$$x - \frac{\pi}{6} = \frac{\pi}{6}, \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

$$\therefore x = \frac{2\pi}{6}, \frac{6\pi}{6} = \frac{\pi}{3}, \pi$$

(b) $\sin x + \cos 2x = 1$
 $\sin x + 1 - 2 \sin^2 x = 1 \Rightarrow \sin x - 2 \sin^2 x = 0 \Rightarrow \sin x (1 - 2 \sin x) = 0$
 $\therefore \sin x = 0, \frac{1}{2}$

$$\left. \begin{aligned} x &= n\pi + (-1)^n \sin^{-1}(0) \\ x &= n\pi + (-1)^n \sin^{-1}\left(\frac{1}{2}\right) \end{aligned} \right\}$$

$$\left. \begin{aligned} \sin x &= c, \quad -1 \leq c \leq 1 \\ x &= n\pi + (-1)^n \sin^{-1}(c) \end{aligned} \right\}$$

$$\left. \begin{aligned} x &= n\pi \\ x &= n\pi + (-1)^n \frac{\pi}{6} \end{aligned} \right\}$$

Question 3

(c) (i) $x^2 = 4y \Rightarrow y = \frac{1}{4}x^2$

$$\frac{dy}{dx} = \frac{x}{2} \Rightarrow \frac{dy}{dx_{x=2p}} = \frac{2p}{2} = p$$

$$\therefore m_1 = -\frac{1}{p} \quad \boxed{m_1 \text{ is the gradient of the normal}}$$

$$\therefore y - p^2 = -\frac{1}{p}(x - 2p) \Rightarrow py - p^3 = -x + 2p \Rightarrow x + py = 2p + p^3$$

(ii) $(-2, 5)$ lies on the normal.

$$(-2) + p(5) = 2p + p^3 \Rightarrow p^3 - 3p + 2 = 0$$

Let $P(x) = x^3 - 3x + 2$

$$P(1) = 0 \Rightarrow (x - 1) \text{ is a factor of } P(x).$$

$$x^2 + x - 2 = (x - 1)(x + 2)$$

$$\begin{array}{r} x^2 + x - 2 \\ x-1 \overline{) x^3 + 2} \\ \underline{x^3 - x^2} \\ x^2 - 3x + 2 \\ x-1 \overline{) x^2 - 3x + 2} \\ \underline{x^2 - x} \\ -2x + 2 \\ x-1 \overline{) -2x + 2} \\ \underline{-2x + 2} \\ 0 \end{array}$$

$$\therefore P(x) = (x - 1)^2(x + 2)$$

$$\therefore p^3 - 3p + 2 = 0 \Rightarrow (p - 1)^2(p + 2) = 0$$

$$\therefore p = 1, -2$$

(d) $\deg(A(x)) = 2$ and $\deg(R(x)) < \deg(A(x)) = 2$

(The degree of the remainder is always less than the degree of the divisor).

So the degree of $R(x)$ is at most 1 ie $R(x) = mx + b$ is the most general form.

(e) (i) $\text{LHS} = 2 \times 2 - 7 + 3 = 4 - 7 + 3 = 0 = \text{RHS}$

So $(2, 7)$ lies on $2x - y + 3 = 0$

(ii) $d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}, Ax + By - C = 0 \Leftrightarrow 2x - y - 11 = 0$

$(x_1, y_1) = (2, 7)$

$$d = \frac{|2 \times 2 - 7 - 11|}{\sqrt{2^2 + (-1)^2}} = \frac{|4 - 7 - 11|}{\sqrt{5}} = \frac{14}{\sqrt{5}} = \frac{14\sqrt{5}}{5} \approx 6.261$$

Question 3

(f) $A(-2, -5), B(1, 4)$

$$m_{AB} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 + 5}{1 + 2} = 3 = m_1$$

$$x + 2y + 1 = 0 \Rightarrow m_2 = -\frac{1}{2}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{3 - (-\frac{1}{2})}{1 + 3(-\frac{1}{2})} \right| = \left| \frac{\frac{7}{2}}{-\frac{1}{2}} \right| = 7$$

$$\therefore \theta = 81^\circ 52'$$

(g) (i) $MD = \frac{1}{2} BC = \frac{1}{2} \times \sqrt{4^2 + 4^2} = \frac{1}{2} \times \sqrt{32} = \frac{1}{2} \times 4\sqrt{2} = 2\sqrt{2}$

(ii) $\tan \theta = \frac{AD}{MD} = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}} \Rightarrow \theta = 35^\circ 16'$

QUESTION 4

(a)

$$(i) \frac{-(-4)}{1} = 4 \quad 1$$

$$(ii) \frac{-(-3)}{1} = 3 \quad 1$$

$$(iii) \quad 2 \quad 1$$

$$(iv) = \frac{\beta\gamma\delta + \alpha\gamma\delta + \alpha\beta\delta + \alpha\beta\gamma}{\alpha\beta\gamma\delta} \quad 2$$

$$= \frac{3}{2}$$

b) (i) $6(1)^4 - 7(1)^3 + c(1)^2 + 13(1) - 4 = 0$

$$\Rightarrow c = -8 \quad 2$$

(ii) $6x^4 - 7x^3 - 8x^2 + 13x - 4 = 0$

$$\begin{array}{r} 6x^2 + 5x - 4 \\ x^2 - 2x + 1 \end{array} \Bigg) p(x) \quad 2$$

Other roots are solutions of $6x^2 + 5x - 4 = 0$ i.e. $x = -\frac{4}{3}$ or $x = \frac{1}{2}$

(c) $\log_{10} 2 = \frac{\log_5 2}{\log_5 10}$ (Change of Base)

$$= \frac{\log_5 2}{\log_5 2 + \log_5 5} \quad 3$$

Let $u = \frac{\log_5 2}{\log_5 2 + 1}$

(Since $\log_5 8 = a \Rightarrow 3\log_5 2 = a$)
(1.1)

$$\therefore u = \frac{\frac{a}{3}}{\frac{a}{3} + 1} = \frac{a}{a+3}$$

(d)

(i) $\cos 3A = \cos(2A+A) \quad 3$

$$= \cos 2A \cos A - \sin 2A \sin A$$

$$= (2\cos^2 A - 1)\cos A - 2\sin A \cos A \sin A$$

$$= 2\cos^3 A - \cos A - 2\cos A [1 - \cos^2 A]$$

$$= 4\cos^3 A - 3\cos A$$

(ii)

$$2\cos 3A = 8\cos^3 A - 6\cos A$$

$$= (2\cos A)^3 - 3(2\cos A)$$

$$= \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right)$$

$$= \dots \text{etc.}$$

$$= x^3 + \frac{1}{x^3} \quad 2$$

e) Bookwork

(i) $y = qx - aq^2$ 1

(ii) $M \left[a(p+q), \frac{a}{2}(p^2+q^2) \right]$ 1

(iv) $T \left[a(p+q), apq \right]$ 2

(v) gradient of TM
$$\frac{\frac{a}{2}(p^2+q^2) - apq}{0}$$
 2

$\Rightarrow$ undefined gradient.

and gradient of axis of parabola
is undefined

$\therefore$ TM // axis

(vi) $K \left[a(p+q), \frac{\frac{a}{2}(p^2+q^2) + apq}{2} \right]$ 2

let $x = a(p+q)$, $y = \frac{a}{4}(p^2+q^2) + \frac{apq}{2}$

$$= \frac{a}{4} [p^2 + q^2 + 2pq]$$

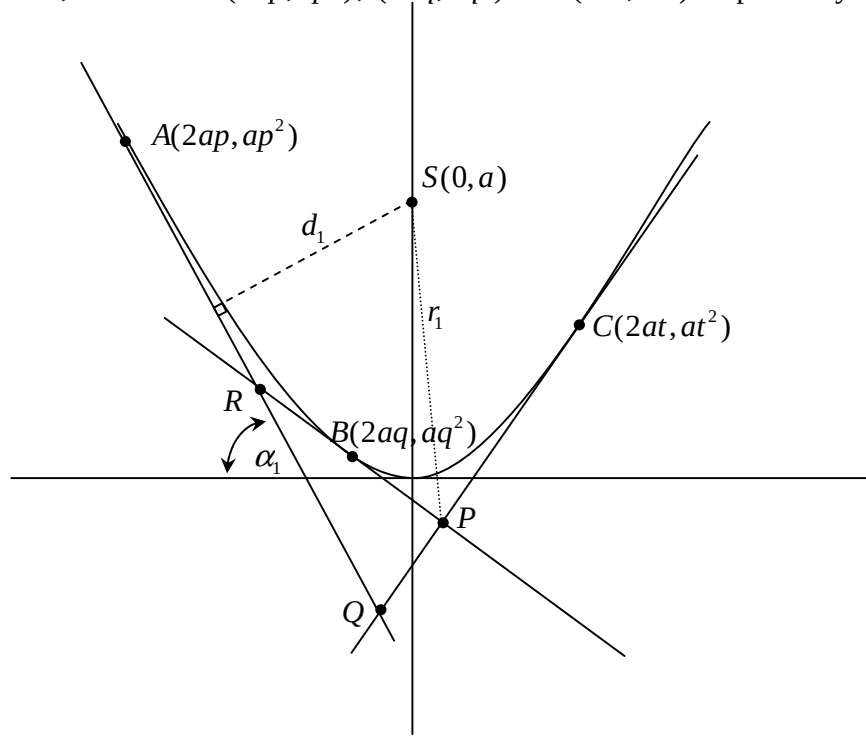
$$= \frac{a}{4} [p+q]^2$$

$$= \frac{a}{4} \left[\frac{x}{a} \right]^2$$

$$y = \frac{x^2}{4a} \text{ locus}$$

Question 4 (f)

Let the points A, B and C be $(2ap, ap^2)$, $(2aq, aq^2)$ and $(2at, at^2)$ respectively.



The gradient of AR is p , so that $\tan \alpha_1 = |p|$ [$\because \alpha_1 < 90^\circ$]

So the equation of RQ is $y = px - ap^2 \Leftrightarrow px - y - ap^2 = 0$.

Similarly, the equations of PQ and RP are respectively $y = tx - at^2$ & $y = qx - aq^2$

By solving simultaneously ie the intersection of lines RBP and CPQ , P has coordinates $(a(t+q), atq)$. This was proved in 4(e).

(Similarly Q and R have coordinates $(a(t+p), atp)$ & $(a(t+q), atq)$ respectively)

If $\tan \alpha_1 = |p| \Rightarrow \cos \alpha_1 = \frac{1}{\sqrt{1+p^2}}$ [Pythagoras' Theorem]

(Similarly $\cos \alpha_2 = \frac{1}{\sqrt{1+q^2}}$ & $\cos \alpha_3 = \frac{1}{\sqrt{1+t^2}}$)

$$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}} \Rightarrow d_1 = \frac{a(1+p^2)}{\sqrt{1+p^2}} = a\sqrt{1+p^2}$$

(Similarly $d_2 = a\sqrt{1+q^2}$ & $d_3 = a\sqrt{1+t^2}$)

$$r_1^2 = PS^2 = (a^2(t+q)^2 + a^2(tq-1)^2) = a^2(t^2 + q^2 + 1 + t^2q^2) = a^2(1+t^2)(1+q^2)$$

(Similarly $r_2^2 = a^2(1+p^2)(1+t^2)$ & $r_3^2 = a^2(1+p^2)(1+q^2)$)

$$(i) \quad d_1 \cos \alpha_1 = a \sqrt{1+p^2} \times \frac{1}{\sqrt{1+p^2}} = a.$$

Similarly for $d_2 \cos \alpha_2$ & $d_3 \cos \alpha_3$ ie $d_2 \cos \alpha_2 = d_3 \cos \alpha_3 = a$

QED

$$(ii) \quad d_1^2 r_1^2 = a^2(1+p^2) \times a^2(1+q^2)(1+t^2) = a^4(1+p^2)(1+q^2)(1+t^2)$$

Similarly $d_2^2 r_2^2 = d_3^2 r_3^2 = a^4(1+p^2)(1+q^2)(1+t^2)$

Thus $d_1 r_1 = d_2 r_2 = d_3 r_3$

QED

$$(iii) \quad r_1 r_2 r_3 = \frac{d_1^2 d_2^2 d_3^2}{a^3}$$

$$\Leftrightarrow a^3 r_1 r_2 r_3 = d_1^2 d_2^2 d_3^2$$

$$\Leftrightarrow (a r_1)(a r_2)(a r_3) = d_1^2 d_2^2 d_3^2$$

$$\Leftrightarrow (d_1 r_1 \cos \alpha_1)(d_2 r_2 \cos \alpha_2)(d_3 r_3 \cos \alpha_3) = d_1^2 d_2^2 d_3^2 \quad (\text{from (i)})$$

$$\Leftrightarrow (r_1 \cos \alpha_1)(r_2 \cos \alpha_2)(r_3 \cos \alpha_3) = d_1 d_2 d_3$$

$$(r_1^2 \cos^2 \alpha_1)(r_2^2 \cos^2 \alpha_2)(r_3^2 \cos^2 \alpha_3)$$

$$= \frac{a^2(1+q^2)(1+t^2)}{(1+p^2)} \times \frac{a^2(1+p^2)(1+t^2)}{(1+q^2)} \times \frac{a^2(1+q^2)(1+p^2)}{(1+t^2)}$$

$$= a^6(1+q^2)(1+t^2)(1+p^2)$$

$$= a^2(1+p^2) \times a^2(1+q^2) \times a^2(1+t^2)$$

$$= d_1^2 d_2^2 d_3^2$$

$$\therefore (r_1 \cos \alpha_1)(r_2 \cos^2 \alpha_2)(r_3 \cos \alpha_3) = d_1 d_2 d_3$$

QED